

Predator Mouths — Teeth Inside Animated Body Geometry

Twelve locked comparisons repair the exact dentures failure: sharks now lift and advance an actual upper-head envelope containing the tooth-bearing jaw, while the orca's fixed rostrum opens against a white chin cut from its real hull. Great white and orca cover rest, partial, full profile, prey-view and contact/head-on states; bull shark and megalodon prove the shared contract generalizes.

12

MATCHED SUBJECTS

1

DEVICE FRAMES

24

BROWSER SCREENSHOTS

Frames: 1200x760

BEFORE · DEPLOYED DENTURE / STATIC-FACE GEOMETRY

<https://efoltyn.github.io/gta6/?visualCompare=before-1787371842877>

AFTER · ARTICULATED BODY ENVELOPES

<http://127.0.0.1:8766/?visualCompare=after-1787371844679>

Generated August 22, 2026 at 12:10 AM. Both columns execute the same local staging recipe through registered production species builders, CBZ.buildSwimRig and CBZ.swimJaw. The before source is the deployed build; the runner copies each deployed camera into the local after capture. No model is reconstructed in this preset.

Σ

Measurements

Species anatomy decides upper travel: sharks lift/protrude the upper envelope; an orca correctly leaves its rostrum fixed. Static chin depth is the failure metric and should reach zero; lower-envelope travel proves body geometry, not loose teeth, opens.

SUBJECT	METRIC	BEFORE	AFTER	Δ
Great White — Sealed Profile, Great White — Opening Profile, Great White — Full Gape Profile, Great White — Full Gape Three-Quarter, Great White — Tuna Inside the Gape, Bull Shark — Shared Envelope, Megalodon — Shared Envelope at Scale	Static body left below the moving jaw (dentures failure) _m	0	0	0
Great White — Sealed Profile, Orca — Sealed Profile, Orca — Opening Profile, Orca — Full Gape Profile, Orca — Full Gape Three-Quarter, Orca — Full Gape Head-On	Actual upper body-envelope travel _m	—	0	0
Great White — Sealed Profile	Actual lower body-envelope travel _m	0	0	0
Great White — Opening Profile	Actual upper body-envelope travel _m	—	0.11	+0.11
Great White — Opening Profile	Actual lower body-envelope travel _m	0.54	0.54	0%
Great White — Full Gape Profile, Great White — Full Gape Three-Quarter	Actual upper body-envelope travel _m	—	0.24	+0.24
Great White — Full Gape Profile, Great White — Full Gape Three-Quarter	Actual lower body-envelope travel _m	1.17	1.17	0%
Great White — Tuna Inside the Gape	Actual upper body-envelope travel _m	—	0.2	+0.2
Great White — Tuna Inside the Gape	Actual lower body-envelope travel _m	0.98	0.98	0%
Orca — Sealed Profile, Orca — Opening Profile, Orca — Full Gape Profile, Orca — Full Gape Three-Quarter, Orca — Full Gape Head-On	Static body left below the moving jaw (dentures failure) _m	0.55	0	-100%
Orca — Sealed Profile	Actual lower body-envelope travel _m	—	0	0
Orca — Opening Profile	Actual lower body-envelope travel _m	—	0.53	+0.53
Orca — Full Gape Profile, Orca — Full Gape Three-Quarter, Orca — Full Gape Head-On	Actual lower body-envelope travel _m	—	1.2	+1.2
Bull Shark — Shared Envelope	Actual upper body-envelope travel _m	—	0.15	+0.15
Bull Shark — Shared Envelope	Actual lower body-envelope travel _m	0.77	0.77	0%
Megalodon — Shared Envelope at Scale	Actual upper body-envelope travel _m	—	0.94	+0.94
Megalodon — Shared Envelope at Scale	Actual lower body-envelope travel _m	4.33	4.33	0%

01 Great White — Sealed Profile

The upper teeth and gums must disappear into the real rostrum envelope; the white chin must close the same body silhouette below it.

BEFORE

BEFORE · DEPLOYED DENTURE / STATIC-FACE GEOMETRY

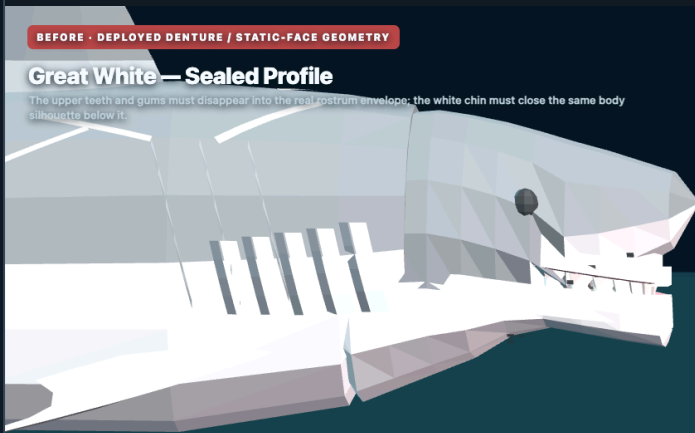
BEFORE · DEPLOYED DENTURE / STATIC-FACE GEOMETRY

REST · BODY SEALED

JAW 0%

Great White — Sealed Profile

The upper teeth and gums must disappear into the real rostrum envelope; the white chin must close the same body silhouette below it.



efoltyn.github.io/gta6/

No white denture rail · no static chin behind jaw

AFTER

AFTER · ARTICULATED BODY ENVELOPES

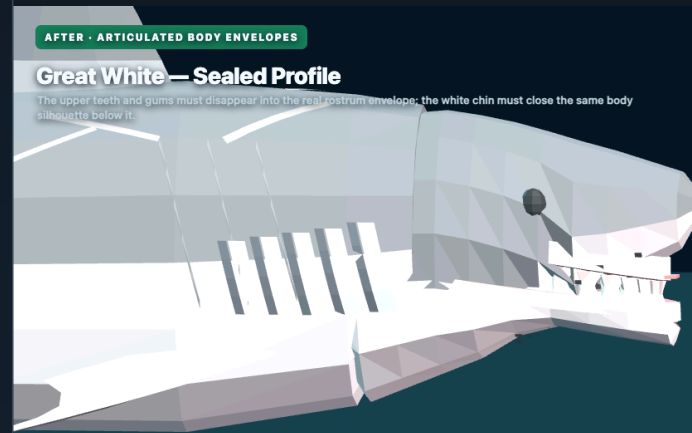
AFTER · ARTICULATED BODY ENVELOPES

REST · BODY SEALED

JAW 0%

Great White — Sealed Profile

The upper teeth and gums must disappear into the real rostrum envelope; the white chin must close the same body silhouette below it.



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No white denture rail · no static chin behind jaw

02 Great White — Opening Profile

At wind-up, real crown vertices lift and advance with the tooth-bearing upper jaw; the teeth remain rooted inside that tissue-bounded envelope.

BEFORE

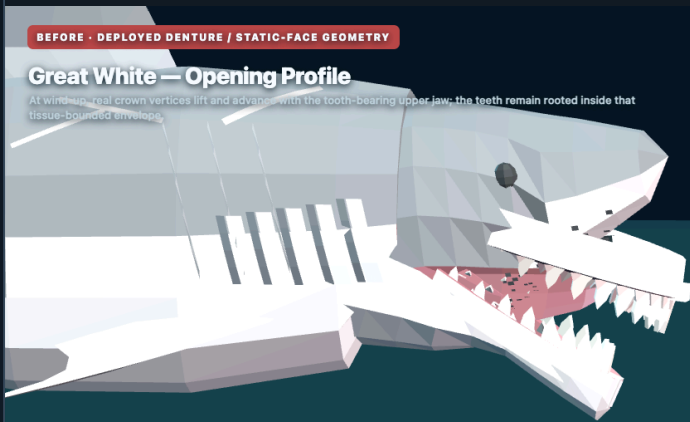
BEFORE · DEPLOYED DENTURE / STATIC-FACE GEOMETRY

BEFORE · DEPLOYED DENTURE / STATIC-FACE GEOMETRY

WIND-UP · 42%
JAW 42%

Great White — Opening Profile

At wind-up, real crown vertices lift and advance with the tooth-bearing upper jaw; the teeth remain rooted inside that tissue-bounded envelope.



efoltn.github.io/gta6/

Upper body envelope advances · teeth stay embedded

AFTER

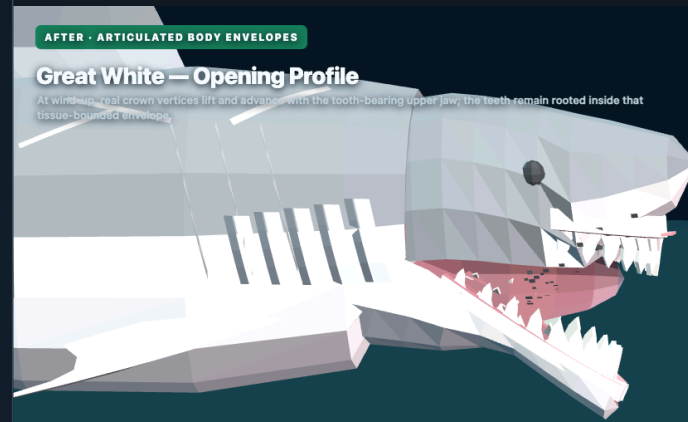
AFTER · ARTICULATED BODY ENVELOPES

AFTER · ARTICULATED BODY ENVELOPES

WIND-UP · 42%
JAW 42%

Great White — Opening Profile

At wind-up, real crown vertices lift and advance with the tooth-bearing upper jaw; the teeth remain rooted inside that tissue-bounded envelope.



127.0.0.1:8766/

Upper body envelope advances · teeth stay embedded

03 Great White — Full Gape Profile

BEFORE

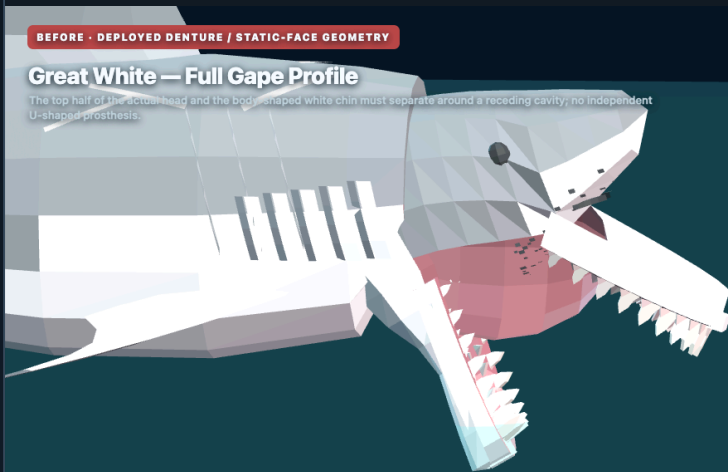
BEFORE · DEPLOYED DENTURE / STATIC-FACE GEOMETRY

BEFORE · DEPLOYED DENTURE / STATIC-FACE GEOMETRY

FULL GAPE · PROFILE
JAW 100%

Great White — Full Gape Profile

The top half of the actual head and the body-shaped white chin must separate around a receding cavity; no independent U-shaped prosthesis.



efoltyn.github.io/gta6/

Actual rostrum travels · actual chin travels

AFTER

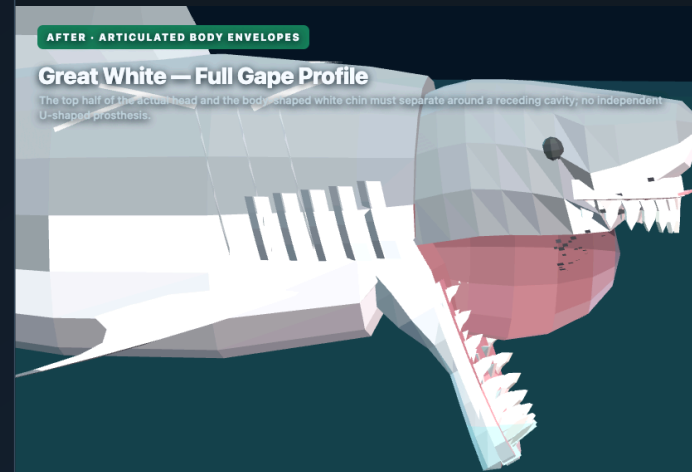
AFTER · ARTICULATED BODY ENVELOPES

AFTER · ARTICULATED BODY ENVELOPES

FULL GAPE · PROFILE
JAW 100%

Great White — Full Gape Profile

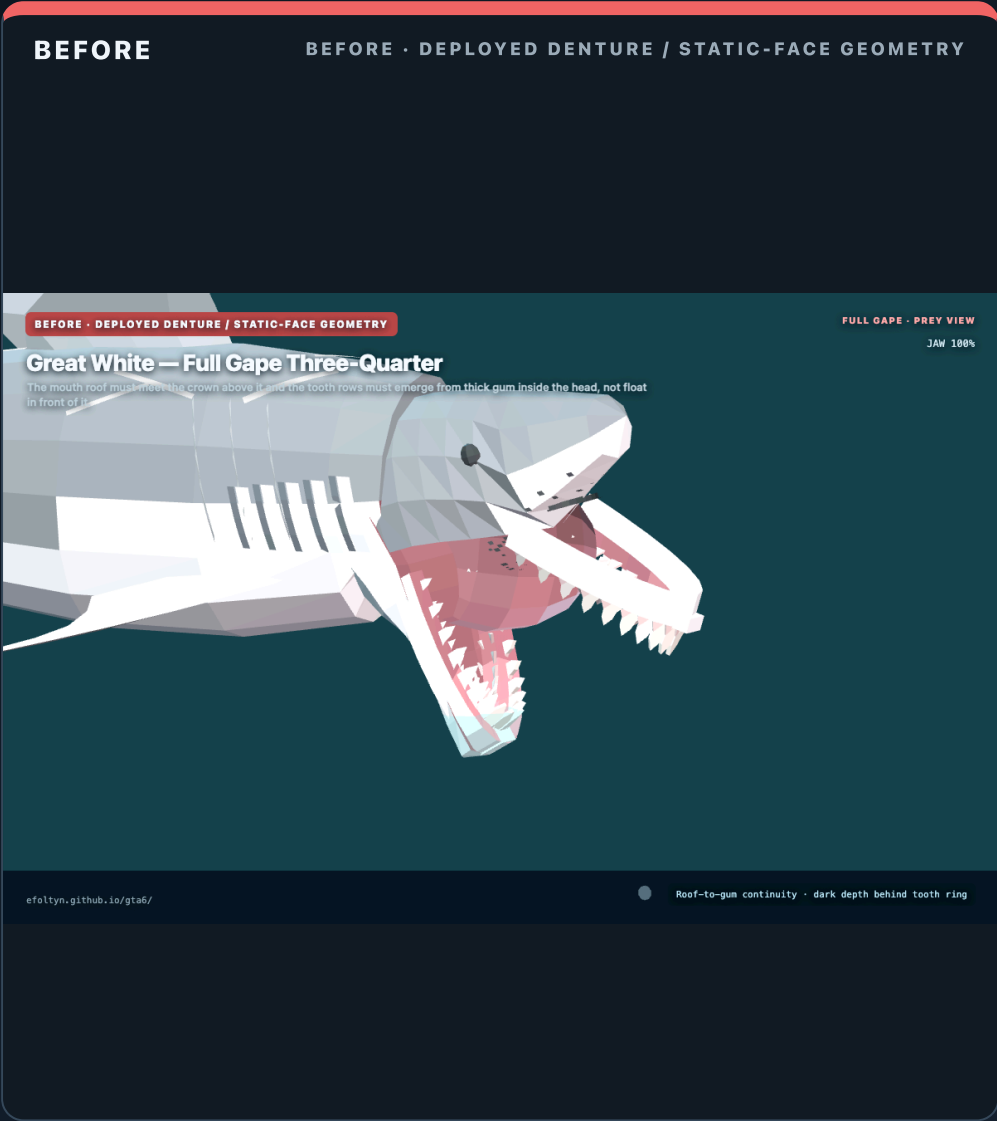
The top half of the actual head and the body-shaped white chin must separate around a receding cavity; no independent U-shaped prosthesis.



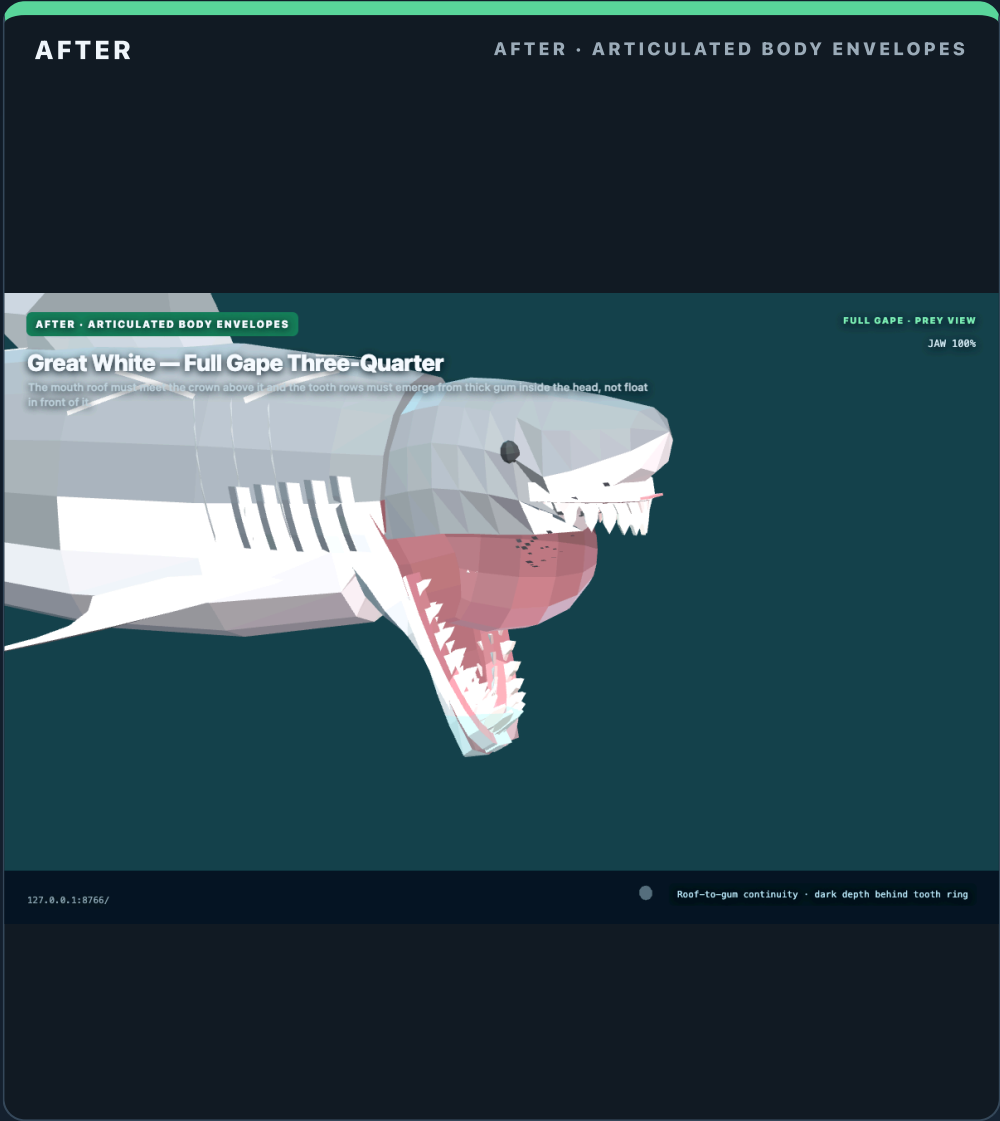
127.0.0.1:8766/

Actual rostrum travels · actual chin travels

04 Great White — Full Gape Three-Quarter



The mouth roof must meet the crown above it and the tooth rows must emerge from thick gum inside the head, not float in front of it.



05 Great White — Tuna Inside the Gape

BEFORE

BEFORE · DEPLOYED DENTURE / STATIC-FACE GEOMETRY

BEFORE · DEPLOYED DENTURE / STATIC-FACE GEOMETRY

Great White — Tuna Inside the Gape

The real tuna must enter between upper and lower body envelopes at the same authored bite socket used for damage.

CONTACT · PREY BETWEEN TEETH

JAW 82%

efoltyn.github.io/gta6/

Target inside mouth, not overlapped by a closed face

AFTER

AFTER · ARTICULATED BODY ENVELOPES

AFTER · ARTICULATED BODY ENVELOPES

Great White — Tuna Inside the Gape

The real tuna must enter between upper and lower body envelopes at the same authored bite socket used for damage.

CONTACT · PREY BETWEEN TEETH

JAW 82%

127.0.0.1:8766/

Target inside mouth, not overlapped by a closed face

06 Orca — Sealed Profile

The white chin must complete the whale's lower-body silhouette and hide the recessed gum roots; only a narrow dark mouth seam remains.

BEFORE

BEFORE · DEPLOYED DENTURE / STATIC-FACE GEOMETRY

BEFORE · DEPLOYED DENTURE / STATIC-FACE GEOMETRY

REST · CHIN IS BODY
JAW 0%

Orca — Sealed Profile

The white chin must complete the whale's lower-body silhouette and hide the recessed gum roots; only a narrow dark mouth seam remains.



efoltyn.github.io/gta6/

Two-thirds of each tooth embedded · no pink stripe

AFTER

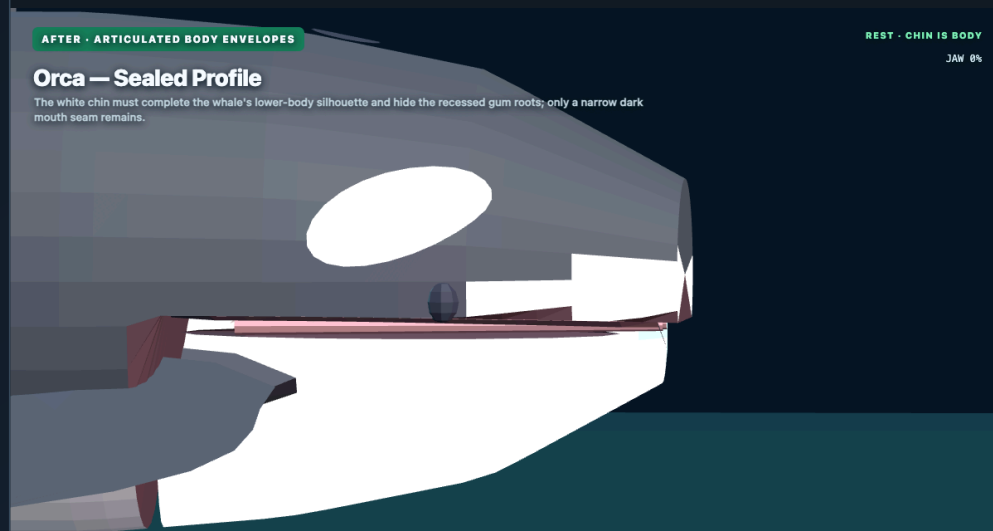
AFTER · ARTICULATED BODY ENVELOPES

AFTER · ARTICULATED BODY ENVELOPES

REST · CHIN IS BODY
JAW 0%

Orca — Sealed Profile

The white chin must complete the whale's lower-body silhouette and hide the recessed gum roots; only a narrow dark mouth seam remains.



127.0.0.1:8766/

Two-thirds of each tooth embedded · no pink stripe

07 Orca — Opening Profile

BEFORE

BEFORE · DEPLOYED DENTURE / STATIC-FACE GEOMETRY

BEFORE · DEPLOYED DENTURE / STATIC-FACE GEOMETRY

OPENING · 45%
JAW 45%

Orca — Opening Profile

An orca does not protrude its upper jaw: the fixed rostrum becomes the roof while the actual white chin rotates from its body hinge.



AFTER

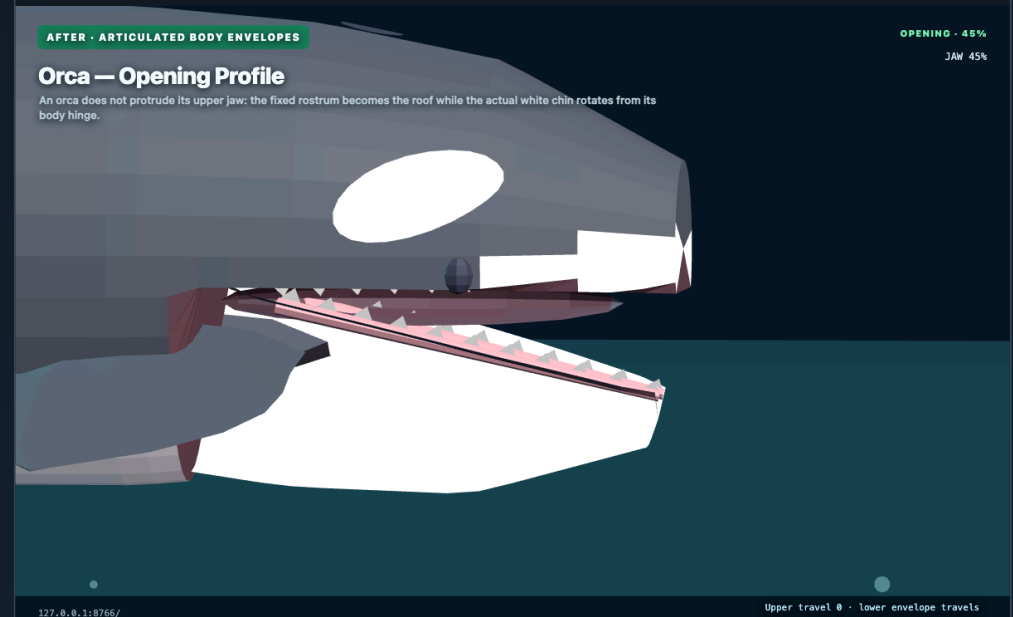
AFTER · ARTICULATED BODY ENVELOPES

AFTER · ARTICULATED BODY ENVELOPES

OPENING · 45%
JAW 45%

Orca — Opening Profile

An orca does not protrude its upper jaw: the fixed rostrum becomes the roof while the actual white chin rotates from its body hinge.



08 Orca — Full Gape Profile

BEFORE

BEFORE · DEPLOYED DENTURE / STATIC-FACE GEOMETRY

BEFORE · DEPLOYED DENTURE / STATIC-FACE GEOMETRY

Orca — Full Gape Profile

The intact lower face must be gone from behind the jaw. The mouth is the black upper body separating from its own white chin.

FULL GAPE · PROFILE
JAW 100%

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Zero static chin depth · one physical mandible

AFTER

AFTER · ARTICULATED BODY ENVELOPES

AFTER · ARTICULATED BODY ENVELOPES

Orca — Full Gape Profile

The intact lower face must be gone from behind the jaw. The mouth is the black upper body separating from its own white chin.

FULL GAPE · PROFILE
JAW 100%

127.0.0.1:8766/

Zero static chin depth · one physical mandible

09 Orca — Full Gape Three-Quarter

BEFORE

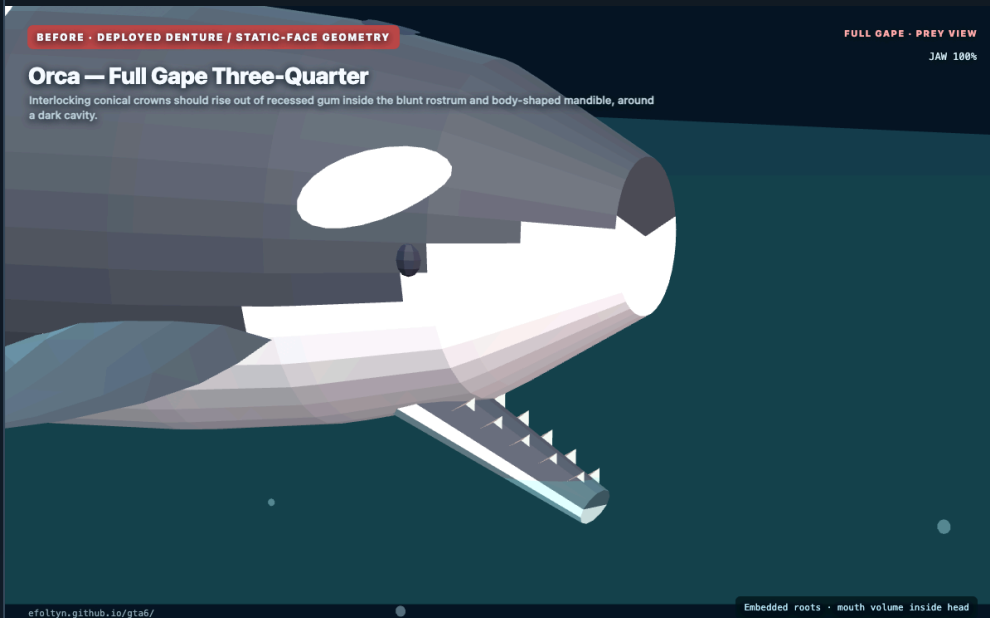
BEFORE · DEPLOYED DENTURE / STATIC-FACE GEOMETRY

BEFORE · DEPLOYED DENTURE / STATIC-FACE GEOMETRY

FULL GAPE · PREY VIEW
JAW 100%

Orca — Full Gape Three-Quarter

Interlocking conical crowns should rise out of recessed gum inside the blunt rostrum and body-shaped mandible, around a dark cavity.



AFTER

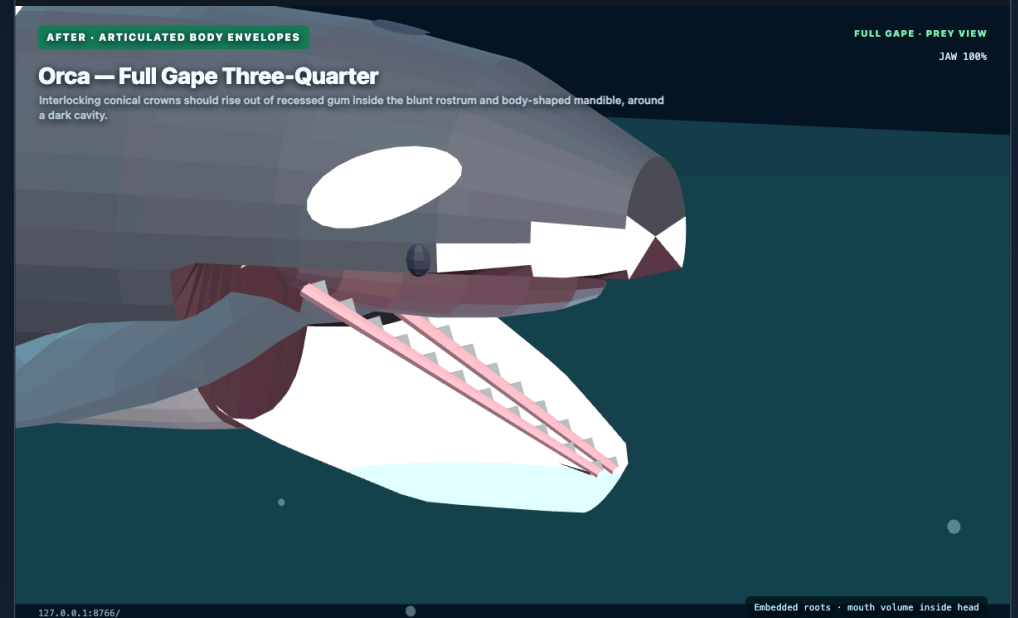
AFTER · ARTICULATED BODY ENVELOPES

AFTER · ARTICULATED BODY ENVELOPES

FULL GAPE · PREY VIEW
JAW 100%

Orca — Full Gape Three-Quarter

Interlocking conical crowns should rise out of recessed gum inside the blunt rostrum and body-shaped mandible, around a dark cavity.



10 Orca — Full Gape Head-On

BEFORE

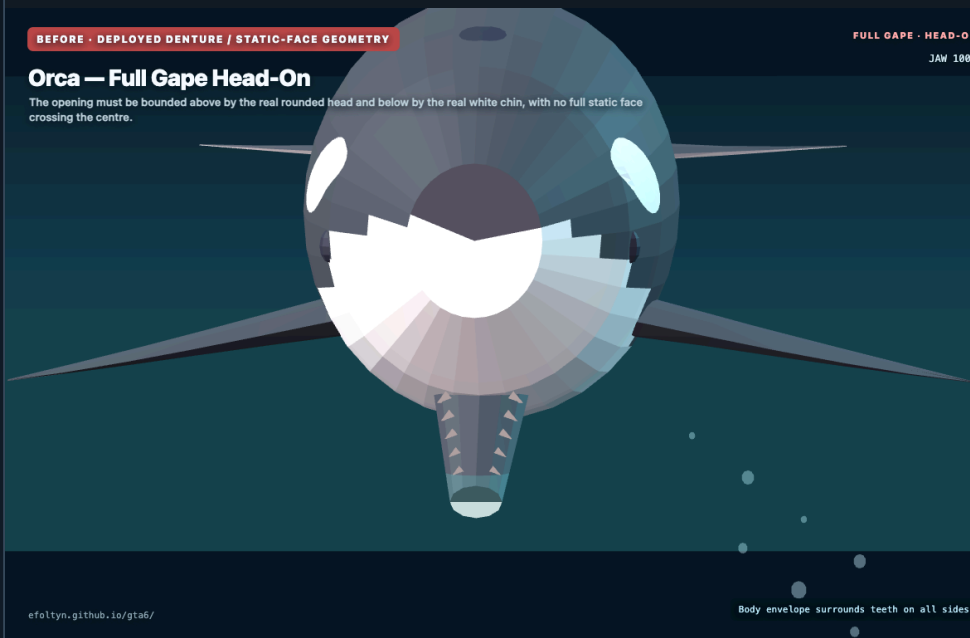
BEFORE · DEPLOYED DENTURE / STATIC-FACE GEOMETRY

BEFORE · DEPLOYED DENTURE / STATIC-FACE GEOMETRY

Orca — Full Gape Head-On

The opening must be bounded above by the real rounded head and below by the real white chin, with no full static face crossing the centre.

FULL GAPE · HEAD-ON
JAW 100%



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Body envelope surrounds teeth on all sides

AFTER

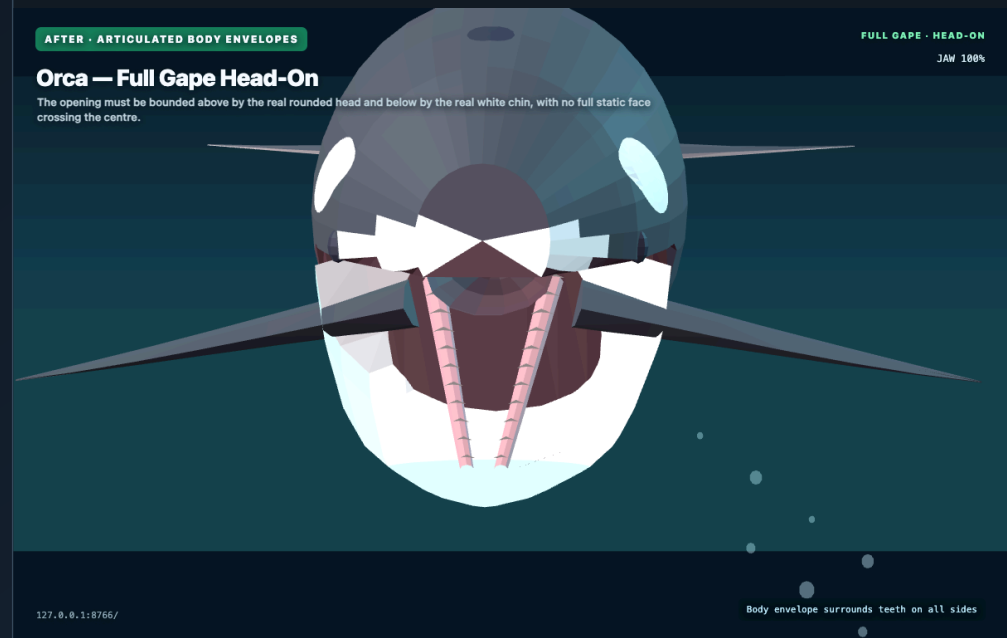
AFTER · ARTICULATED BODY ENVELOPES

AFTER · ARTICULATED BODY ENVELOPES

Orca — Full Gape Head-On

The opening must be bounded above by the real rounded head and below by the real white chin, with no full static face crossing the centre.

FULL GAPE · HEAD-ON
JAW 100%



127.0.0.1:8766/

Body envelope surrounds teeth on all sides

11 Bull Shark — Shared Envelope

The blunt inshore shark inherits the same articulated crown/chin/dentition hierarchy without species-specific runtime animation.

BEFORE

BEFORE · DEPLOYED DENTURE / STATIC-FACE GEOMETRY

BEFORE · DEPLOYED DENTURE / STATIC-FACE GEOMETRY

FULL GAPE · SHARED SHARK GRAMMAR
JAW 100%

Bull Shark — Shared Envelope

The blunt inshore shark inherits the same articulated crown/chin/dentition hierarchy without species-specific runtime animation.



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Contract v4 · real upper and lower shell

AFTER

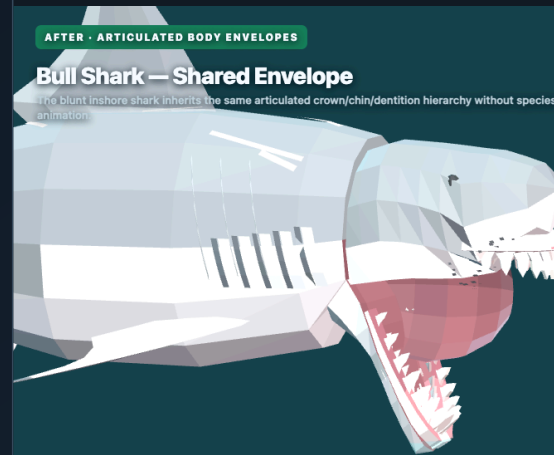
AFTER · ARTICULATED BODY ENVELOPES

AFTER · ARTICULATED BODY ENVELOPES

FULL GAPE · SHARED SHARK GRAMMAR
JAW 100%

Bull Shark — Shared Envelope

The blunt inshore shark inherits the same articulated crown/chin/dentition hierarchy without species-specific runtime animation.



127.0.0.1:8766/

Contract v4 · real upper and lower shell

12 Megalodon — Shared Envelope at Scale

At legendary scale the same mouth hierarchy must remain joined at the cheek and keep every tooth rooted inside moving body geometry.

BEFORE

BEFORE · DEPLOYED DENTURE / STATIC-FACE GEOMETRY

BEFORE · DEPLOYED DENTURE / STATIC-FACE GEOMETRY

FULL GAPE · 2.6x SCALE
JAW 100%

Megalodon — Shared Envelope at Scale

At legendary scale the same mouth hierarchy must remain joined at the cheek and keep every tooth rooted inside moving body geometry.



AFTER

AFTER · ARTICULATED BODY ENVELOPES

AFTER · ARTICULATED BODY ENVELOPES

FULL GAPE · 2.6x SCALE
JAW 100%

Megalodon — Shared Envelope at Scale

At legendary scale the same mouth hierarchy must remain joined at the cheek and keep every tooth rooted inside moving body geometry.

